

COMP-SCI-5565

Intro to Statistical Learning

Topic 7 – Support Vector Machines

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Lecture Objectives

- Implement a binary classification model using a maximal margin classifier.
- Implement a binary classification model using a support vector classifier.
- Implement a binary classification model using a support vector machine (SVM).
- Generalize SVM models to multi-class cases.

Outline

- 7.1 Introduction
- 7.2 Hyperplane
- 7.3 Maximal Margin Classifier
- 7.4 Support Vector Classifier
- 7.5 Nonlinear Classification
- 7.6 Support Vector Machines

Outline

7.1 Introduction

7.2 Hyperplane

7.3 Maximal Margin Classifier

7.4 Support Vector Classifier

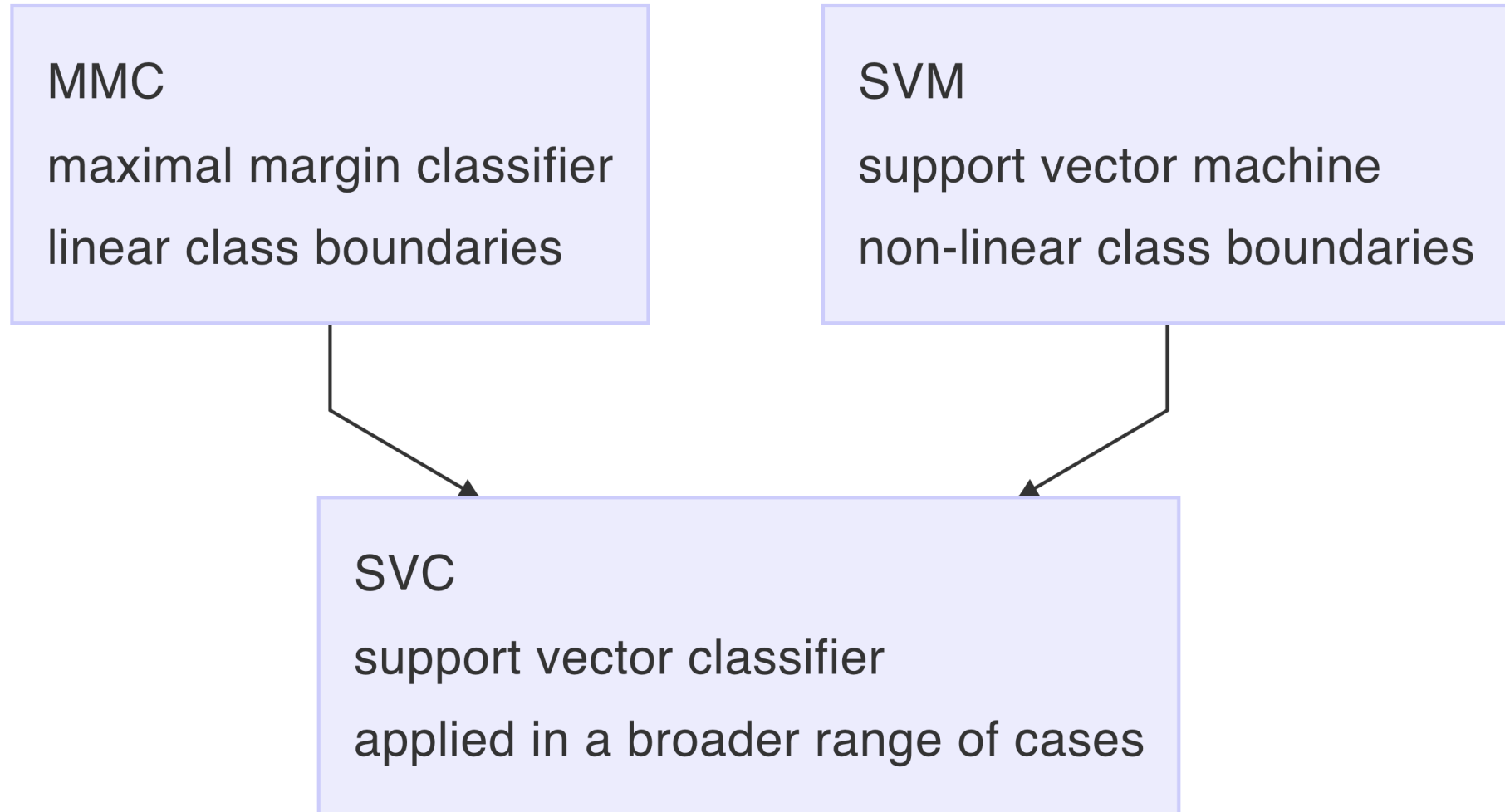
7.5 Nonlinear Classification

7.6 Support Vector Machines

7.1 Introduction

- Here, we directly approach the two-class classification problem:
We try to find a plane that separates the classes in feature space.
- If we cannot, we get creative in two ways:
 - We soften what we mean by “separates” and
 - We enrich and enlarge the feature space so that separation is possible.
- Support vector machine (SVM), an approach for classification, was developed in 1990. SVM is a generalization of classifier methods, in particular:
 - A maximal margin classifier (MMC) requires that the classes be separable by a linear boundary.
 - Support vector classifier (SVC)
 - Support vector machines (SVM)

7.1 Introduction



Outline

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7.2 Hyperplane

7.3 Maximal Margin Classifier

7.4 Support Vector Classifier

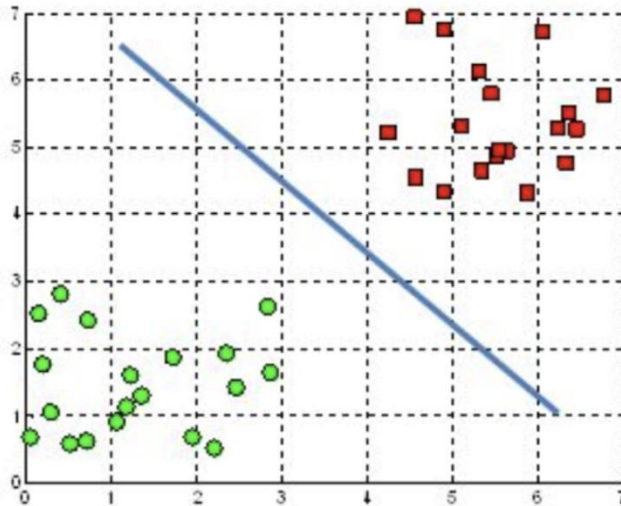
7.5 Nonlinear Classification

7.6 Support Vector Machines

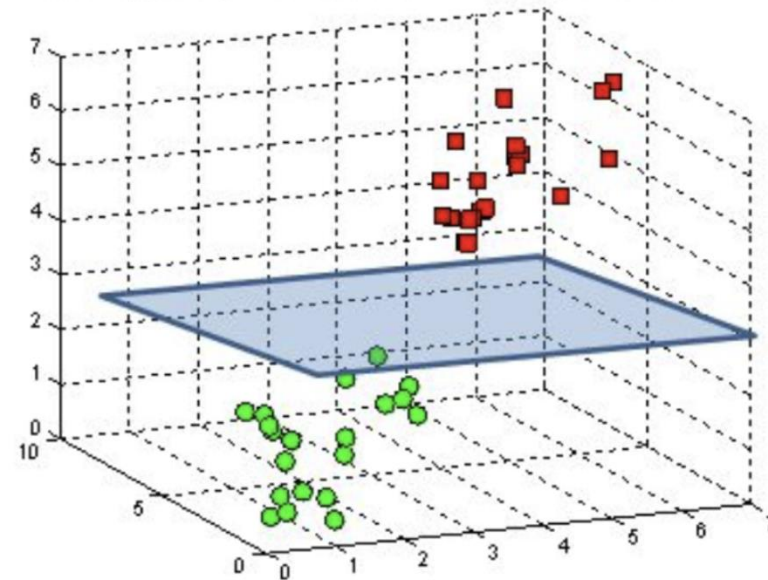
7.2 Hyperplane

- A hyperplane in p dimensions is a flat affine subspace of dimension $p - 1$.
- For example, in a 2-dimensional space, a hyperplane is a flat, one-dimensional space: a line.

A hyperplane in $\mathbb{R}^2$ is a line



A hyperplane in $\mathbb{R}^3$ is a plane



Additional resource: [Deep AI](#)

7.2 Hyperplane

- In general, the equation for a hyperplane has the form

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p = 0$$

- In $p = 2$ dimensions a hyperplane is a line.
- If $\beta_0 = 0$, the hyperplane goes through the origin; otherwise not.
- The vector $\beta = (\beta_1, \beta_2, \dots, \beta_p)$ is called the normal vector – it points in a direction orthogonal to the surface of a hyperplane.

7.2 Hyperplane

Separating Hyperplanes

- Consider a matrix X of dimensions $n \times p$, and a $y_i \in \{-1, 1\}$.
- We have a new observation, x^* , which is a vector $x^* = (x_1^*, \dots, x_p^*)$ which we wish to classify to one of two groups.
- We will use a separating hyperplane to classify the observation.
- A separating hyperplane has the property s.t.

$$\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_p X_{ip} > 0 \text{ if } y_i = 1$$

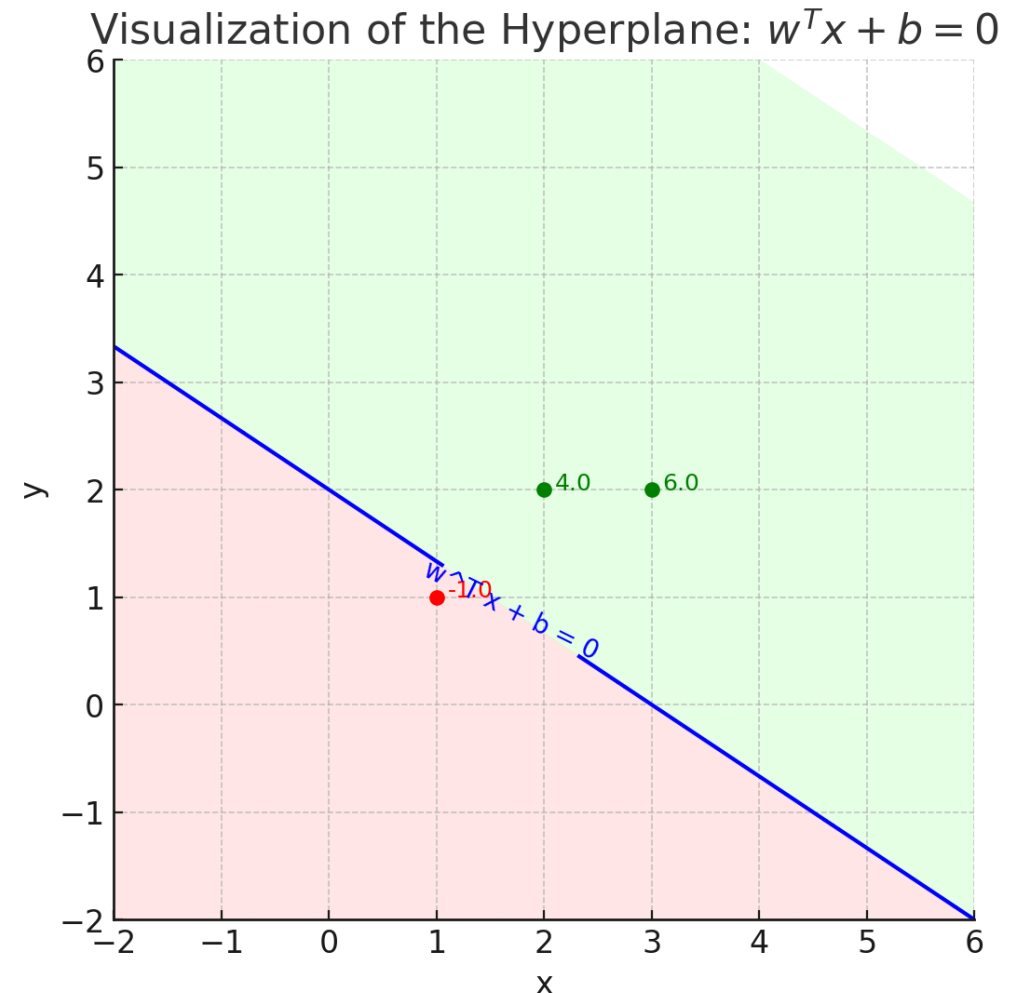
and

$$\beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_p X_{ip} < 0 \text{ if } y_i = -1$$

7.2 Hyperplane

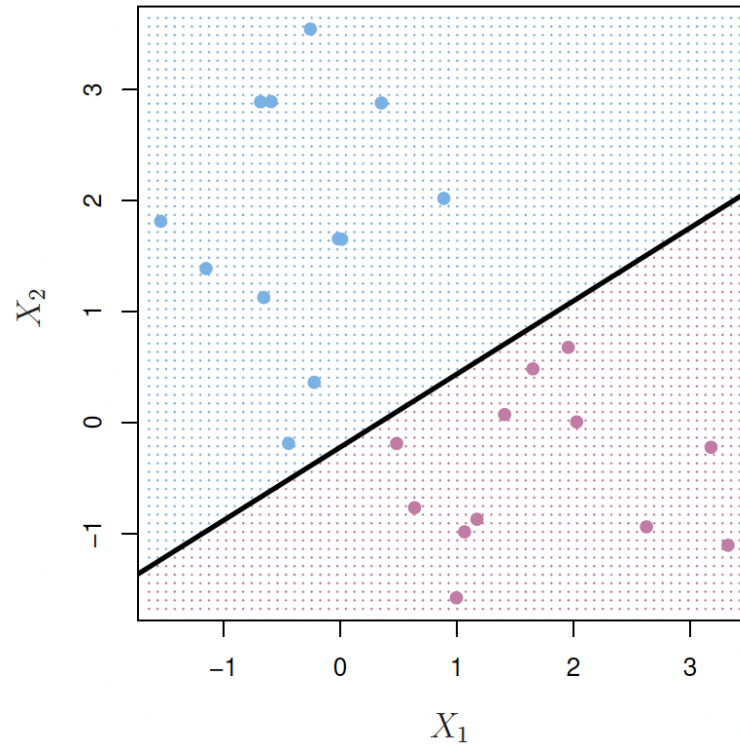
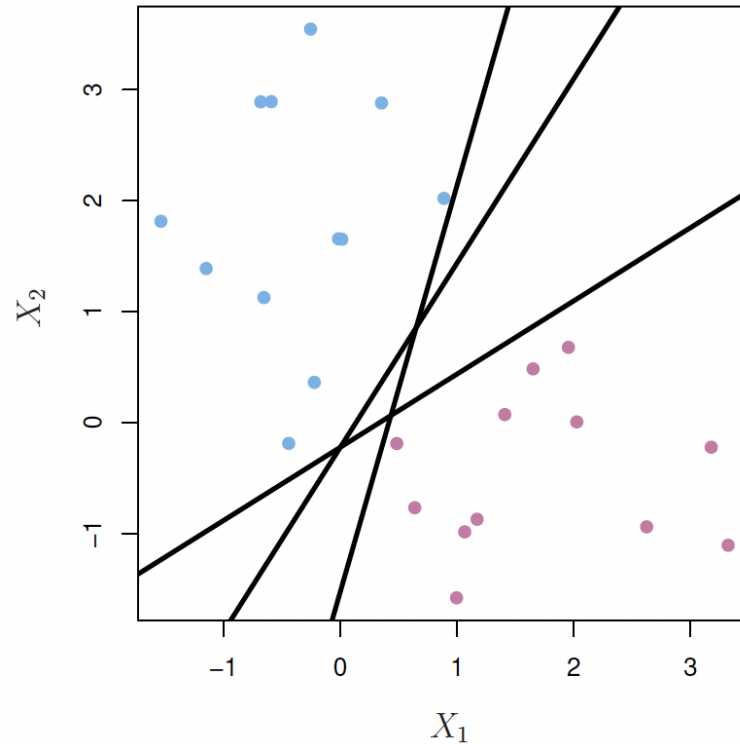
Separating Hyperplanes

- The **blue line** is the hyperplane $W^T x + b = 0$, which defines the boundary.
- The **green region** shows where $W^T x + b > 0$ —points above the hyperplane.
- The **red region** shows where $W^T x + b < 0$ —points below the hyperplane.



7.2 Hyperplane

Separating Hyperplanes



We can label the blue observations as $y_i = 1$ and the pink observations as $y_i = -1$.

Outline

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7.4 Support Vector Classifier

7.5 Nonlinear Classification

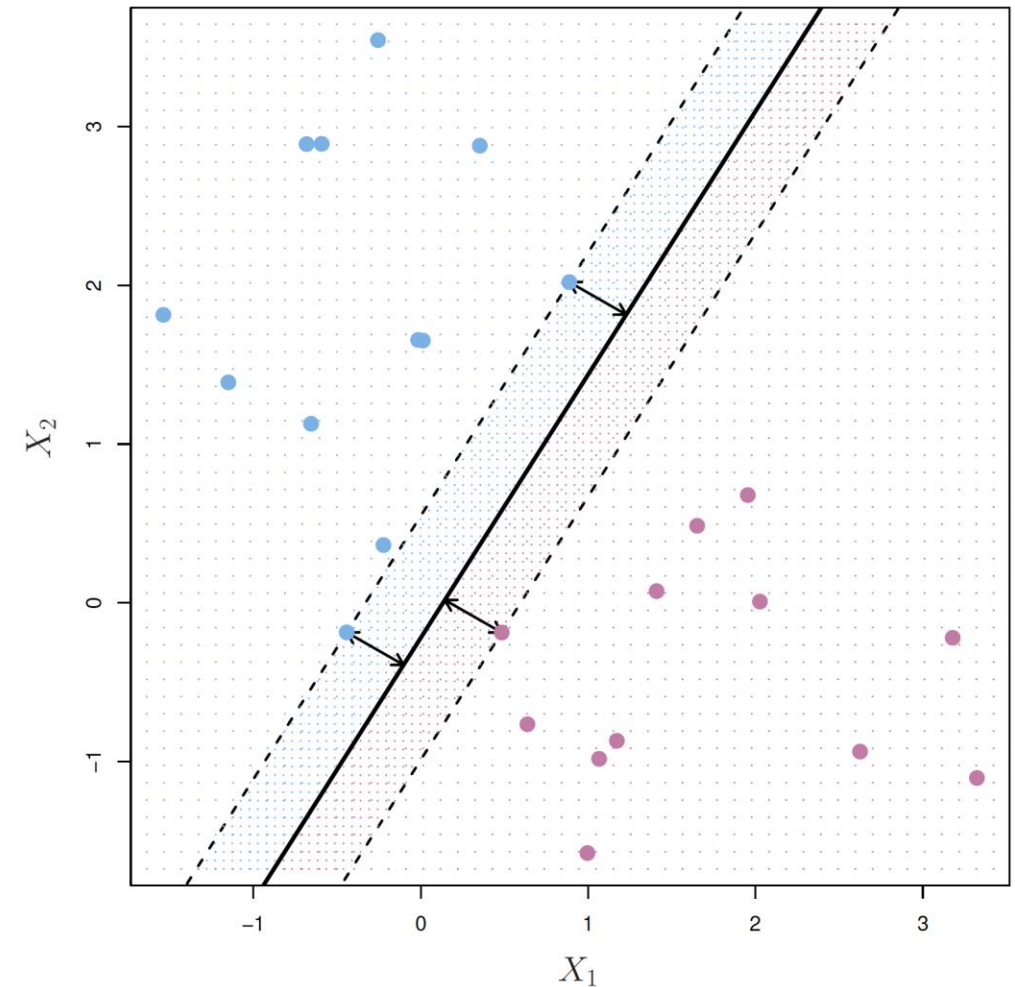
7.6 Support Vector Machines

7.3 Maximal Margin Classifier

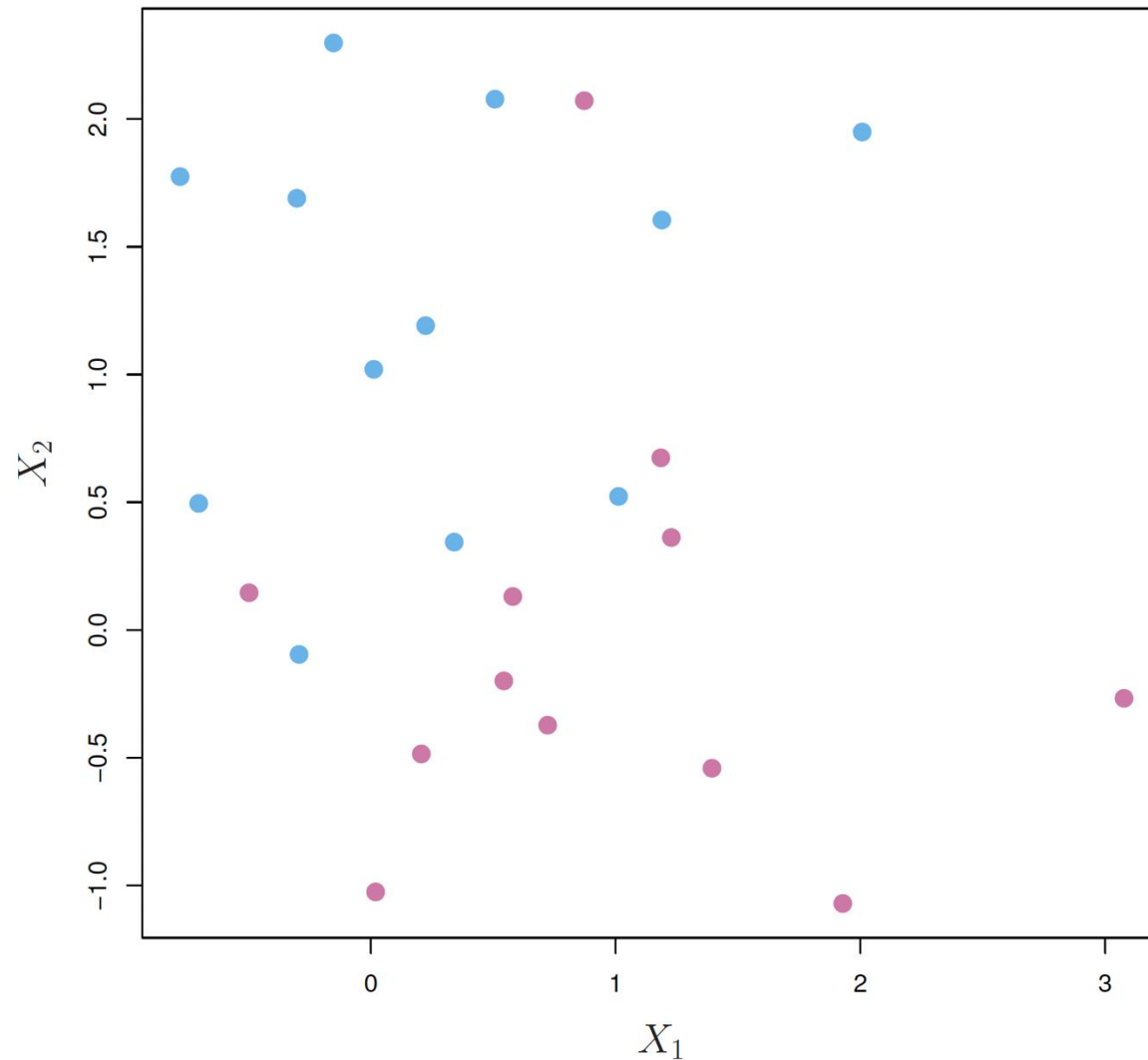
- Generally, if data can be perfectly separated using a hyperplane, an infinite amount of such hyperplanes exist.
- An intuitive choice is the *maximal margin hyperplane*, which is the farthest from the training data.
- We compute the perpendicular distance from each training observation to the hyperplane.
- The smallest of these distances is known as the *margin*.
- The *maximal margin hyperplane* is the hyperplane for which the margin is maximized.
- We can classify a test observation based on which side of the maximal margin hyperplane it lies on, and this is known as the *maximal margin classifier*.

7.3 Maximal Margin Classifier

- Note the 3 training observations that lie on the margin and are equidistant from the hyperplane.
- These are the **support vectors**.
- They support the hyperplane because the hyperplane would change if their location were changed.
- The maximal margin hyperplane depends on these observations *but not the others* (unless the other observations were moved at or within the margin).

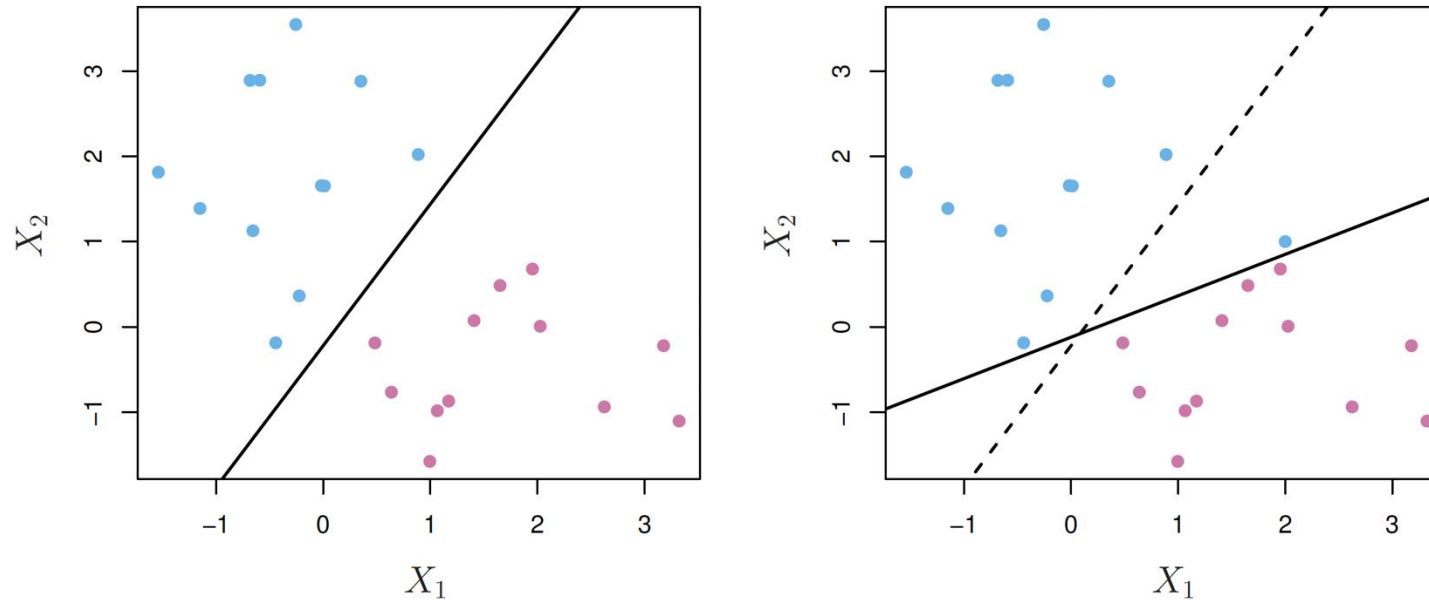


7.3 Maximal Margin Classifier



- But what if our data is not separable by a linear hyperplane?
- The data on the left are not separable by a linear boundary.
- This is often the case

7.3 Maximal Margin Classifier



- Individual data points greatly affect the formation of the maximal margin classifier.
- Sometimes, the data are separable but noisy. This can lead to a poor solution for the maximal-margin classifier.
- The **support vector classifier** maximizes a **soft** margin.

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7.4 Support Vector Classifier

- We often can't always use a hyperplane to separate two classes.
- Even if such a classifier does exist, it's not always desirable due to overfitting or too much sensitivity to individual observations.
- Thus, consider a classifier/hyperplane that misclassifies a few observations to improve the classification of the remaining data points.
- The *support vector classifier (SVC)*, a.k.a the *soft margin classifier*, allows some training data to be on the wrong side of the margin or even the hyperplane.

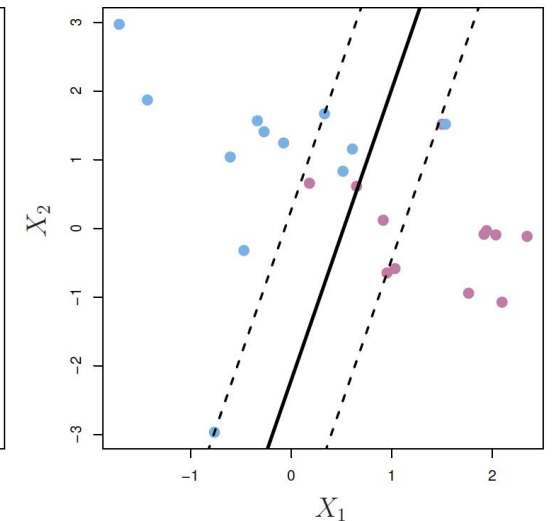
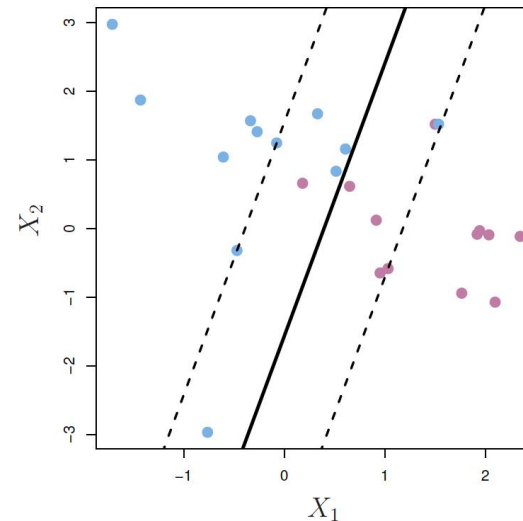
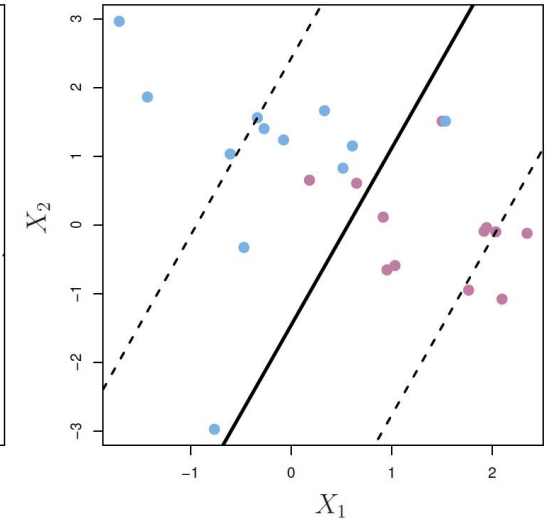
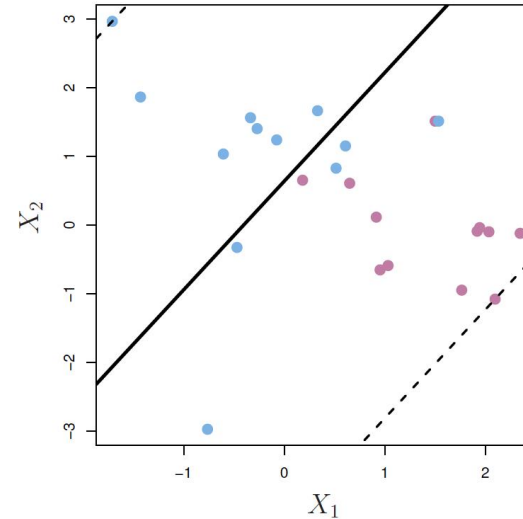
7.4 Support Vector Classifier

- C is a nonnegative tuning parameter, typically chosen through cross-validation, and can be considered as the budget for margin violation by the observations.
- The ϵ_i are *slack variables* that allow individual observations to be on the wrong side of the margin or hyperplane.
- The ϵ_i indicates where the i th observation is located with regards to the margin and hyperplane.
 - If $\epsilon_i = 0$, the observation is on the correct side of the margin.
 - If $\epsilon_i \geq 1$, the observation is on the wrong side of the hyperplane.
- Since C constrains the sum of the ϵ_i , it determines the number and magnitude of violations to the margin. If $C = 0$, there is no margin for violation, thus all the $\epsilon_i, \dots, \epsilon_n = 0$.
- A smaller value of C promotes a wider margin but allows more misclassifications, while a larger value of C prioritizes correct classification at the expense of a narrower margin.

7.4 Support Vector Classifier

C is a regularization parameter

- A property of the classifier is that only data points that lie on or violate the margin will affect the hyperplane.
- These data points are known as *support vectors*.
- C controls the bias-variance tradeoff of the classifier.
 - When C is large: small region of error
 - When C is small: large region of error



Outline

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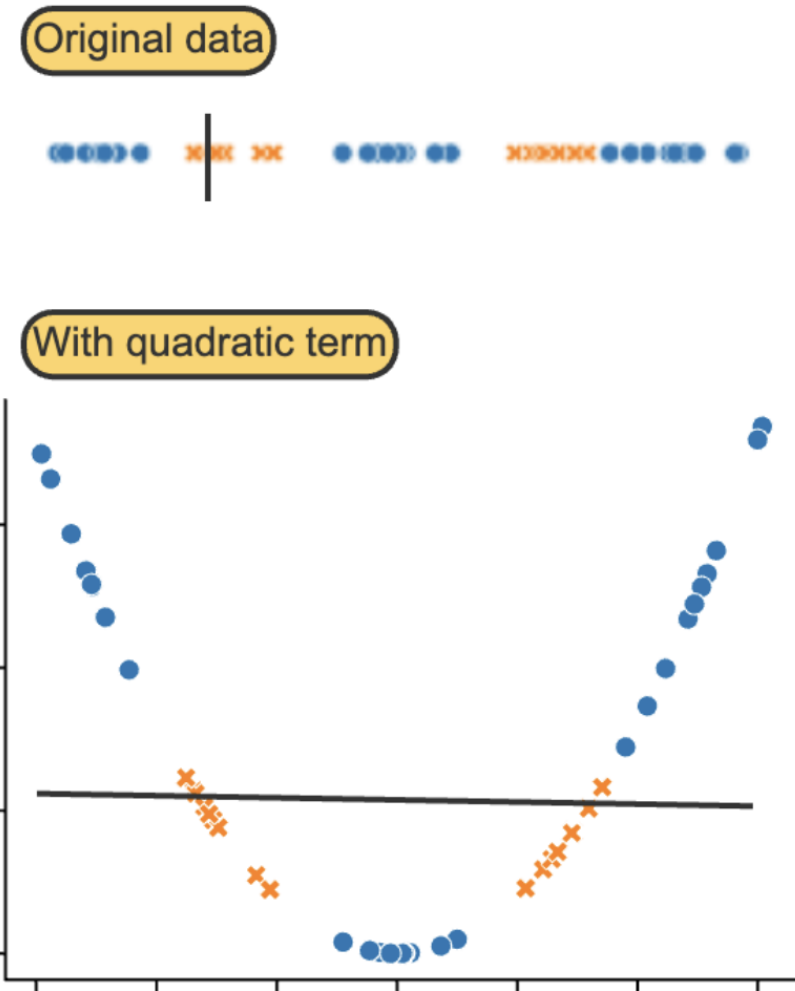
7.4 Support Vector Classifier

7.5 Nonlinear Classification

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7.5 Nonlinear Classification

- Sometimes a linear boundary simply won't work, no matter what value of C .
- The example on the right is such a case.
- Many decision boundaries are not linear.
- What to do?



7.5 Nonlinear Classification

Feature Expansion

- Enlarge the space of features by including transformations; e.g., $X_1^2, X_1^3, X_1X_2, X_1X_2^2, \dots$
- Hence, we go from a p -dimensional space to a $M > p$ dimensional space.
- Fit a support-vector classifier in the enlarged space.
- This results in non-linear decision boundaries in the original space.
- Example: Suppose we use $(X_1, X_2, X_1^2, X_2^2, X_1X_2)$ instead of just (X_1, X_2) . Then the decision boundary would be of the form

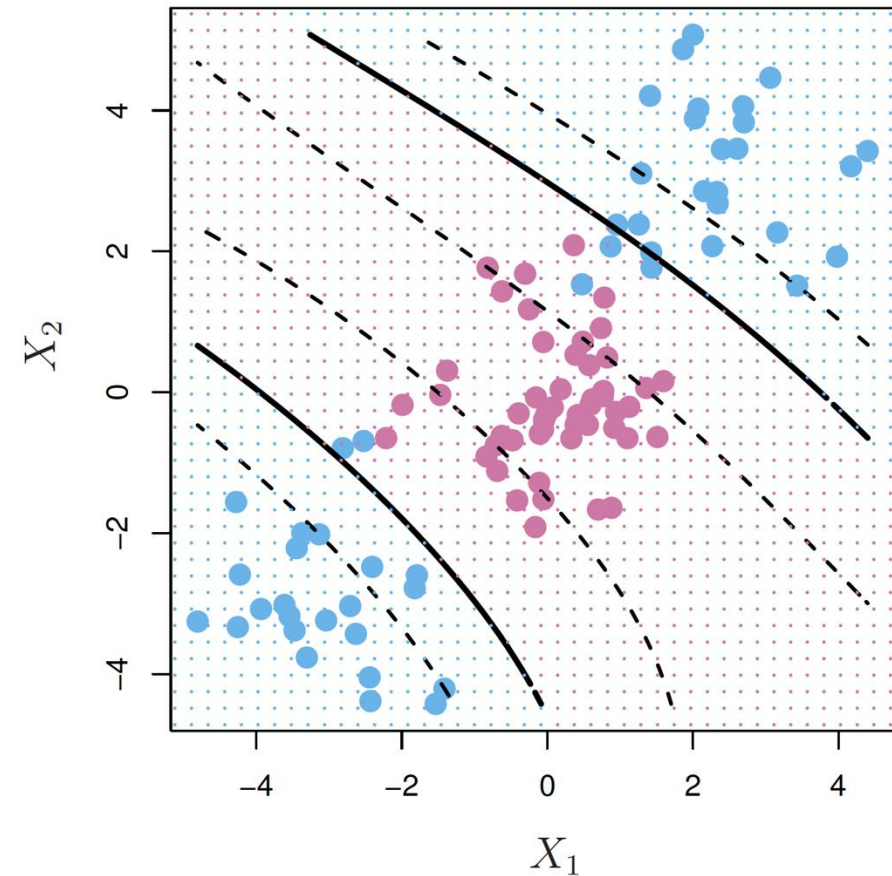
$$\beta_0 + \beta_1X_1 + \beta_2X_2 + \beta_3X_1^2 + \beta_4X_2^2 + \beta_5X_1X_2 = 0$$

- This leads to nonlinear decision boundaries in the original space.

7.5 Nonlinear Classification

Cubic Polynomial

- Here, we use a basis expansion of cubic polynomials.
- From 2 variables to 9.
- The support-vector classifier in the enlarged space solves the problem in the lower-dimensional space.



$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2 + \beta_4 X_2^2 + \beta_5 X_1 X_2 + \beta_6 X_1^3 + \beta_7 X_2^3 + \beta_8 X_1 X_2^2 + \beta_9 X_1^2 X_2 = 0$$

7.5 Nonlinear Classification

Polynomials problem

- Polynomials (especially high-dimensional ones) get wild rather fast – unmanageable calculations.
- There is a more elegant and controlled way to introduce nonlinearities in support-vector classifiers – using kernels.

Outline

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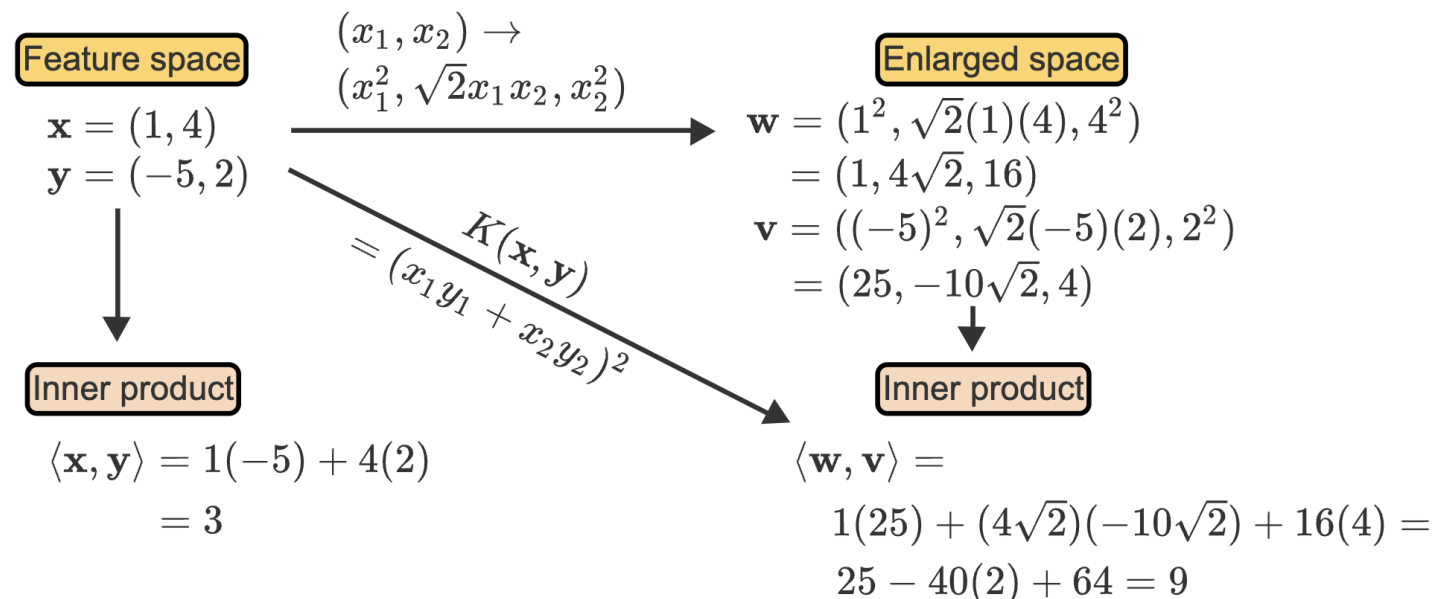
7.4 Support Vector Classifier

7.5 Nonlinear Classification

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7.6 Support Vector Machines

- The Support Vector Machine (SVM) is an SVC extension that results from using *kernels* to enlarge the feature space.
- A **kernel** is a function that quantifies the similarity of two data points.
- The solution to the SVC problem in the SVM context involves only the observations' inner products (AKA dot products). It can be quite abstract!



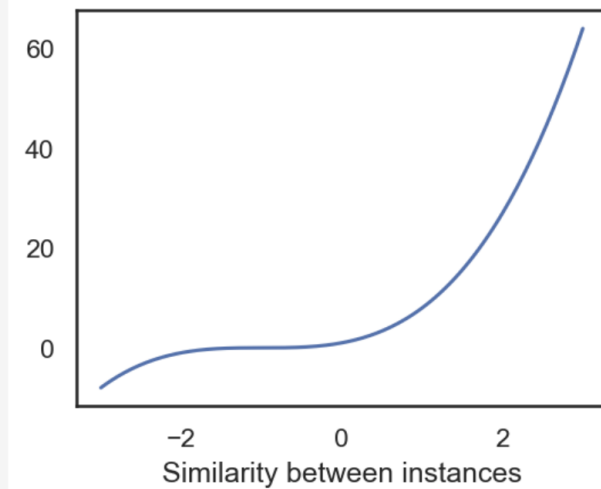
7.6 Support Vector Machines

Commonly used kernels

(1) Polynomial kernel

Polynomial
kernel

$$K(\mathbf{x}, \mathbf{y}) = (1 + \langle \mathbf{x}, \mathbf{y} \rangle)^d$$



Instances are mapped into an enlarged space of polynomial powers similar to the example at the section's start. The decision boundary will curve some.

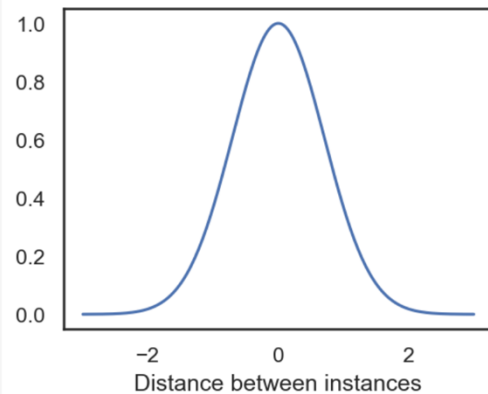
7.6 Support Vector Machines

Commonly used kernels

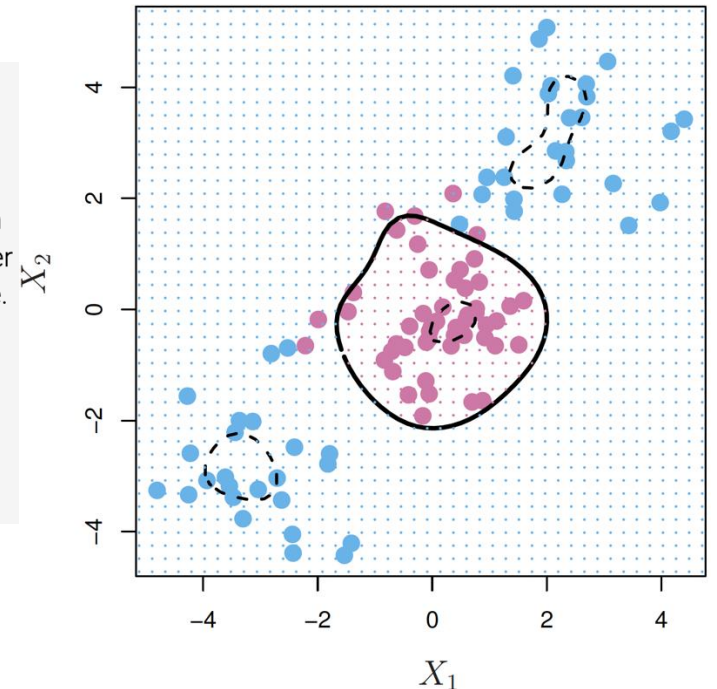
(2) Radial kernel

Radial basis
function
kernel

$$K(\mathbf{x}, \mathbf{y}) = e^{-\gamma|\mathbf{x}-\mathbf{y}|^2}$$



Instances are mapped into an enlarged space where each dimension is related to the distance from one instance. An instance that is closer to a dimension's instance in the feature space will have a larger value. The decision boundary will curve around clusters of instances of the same class.



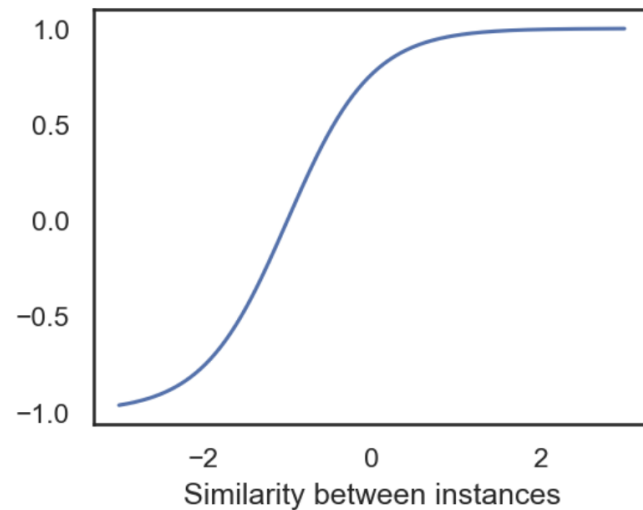
7.6 Support Vector Machines

Commonly used kernels

(3) Sigmoid kernel

Sigmoid
kernel

$$K(\mathbf{x}, \mathbf{y}) = \tanh(\gamma \langle \mathbf{x}, \mathbf{y} \rangle + 1)$$



The sigmoid function is used due to a similar response to biological neurons. The sigmoid function restricts the gains from large inputs to between -1 and 1, reducing the spread of values in the enlarged space.

7.6 Support Vector Machines

More than Two Classes

- The SVM, as defined, works for $K = 2$ classes. What do we do if we have $K > 2$ classes?
- **One versus All (OVA)**
 - Fit K different 2-class SVM classifiers $\hat{f}_k(x)$, $k = 1, \dots, K$; each class versus the rest.
 - Classify x^* to the class for which $\hat{f}_k(x^*)$, is largest.
 - The data point is classified to the class for which it was most often assigned in the pairwise classifications.
- **One versus One (OVO)**
 - Fit all pairwise classifiers $\hat{f}_{k\ell}(x)$.
 - Classify x^* to the class that wins the most pairwise competitions.
 - This can be useful when there are a lot of classes.
- Which to choose? If K is not too large, use OVO.

7.6 Support Vector Machines

Which to use: SVM or Logistic Regression

- When classes are (nearly) separable, SVM does better than LR. So does LDA.
- LR (with ridge penalty) and SVM are very similar when classes are not separable.
- If you wish to estimate probabilities, LR is the choice.
- For nonlinear boundaries, kernel SVMs are popular.
- Can use kernels with LR and LDA, but computations are more expensive.

Summary

- The maximal margin classifier is a separating hyperplane farthest from the training observations.
- The support vector classifier is an extension of the maximal margin classifier, which can be applied in a broader range of cases.
- The support vector machine further extends the support vector classifier, which can accommodate non-linear class boundaries.
- Support vector classifiers and logistic regression often give similar results.
- However, support vector machines perform better when the classes are well separated.
- In cases with more overlap, logistic regression tends to perform better.
- In any case, both should always be tested, and the best method should be chosen.

Further Reading

- Chapter 9 – James G., Witten D., Hastie T., Tibshirani R., & Taylor J. (2023) *An Introduction to Statistical Learning with Applications in Python*. Springer
<https://link.springer.com/book/10.1007/978-3-031-38747-0>
- Chapter 9 - Introduction to Statistical Learning Using R Book Club:
<https://r4ds.github.io/bookclub-islr/support-vector-machines.html>
- Summary of Chapter 9: <https://www.bijenpatel.com/guide/islr/support-vector-machines/>

End of Topic

Thank you
Any questions?